

LAB ASSIGNMENT – 1.5

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TASK – 1:

AI-Generated Logic Without Modularization (String Reversal Without Functions)

Prompt :

Generate a code for AI-Generated Logic Without Modularization (String Reversal Without Functions)

Code:

```
C: > Users > Shree priya > Downloads > aac1.py > ...  
1  # String Reversal Without Functions  
2  input_string = "Hello, World!"  
3  reversed_string = ""  
4  for char in input_string:  
5      reversed_string = char + reversed_string  
6  print(reversed_string)  
7  
8
```

Output:

```
PS C:\Users\Shree priya\Downloads> & "c:\Users\Shree priya\AppData\Local\Microsoft\WindowsApps\python3.13.exe" "c:\Users\Shree priya\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher" "58576" "--" "C:\Users\Shree priya\Downloads\aac1.py"  
IdleW ,olleH  
PS C:\Users\Shree priya\Downloads> ^C  
PS C:\Users\Shree priya\Downloads>
```

TASK – 2:Efficiency & Logic Optimization (Readability Improvement)

Prompt:

Generate a code for Efficiency & Logic Optimization (Readability Improvement)

Code:

```
# Readability Improvement  
input_string = "Hello, World!"  
reversed_string = ""  
for char in input_string:  
    reversed_string = char + reversed_string  
print(reversed_string)
```

Output :

```
PS C:\Users\Shree priya\Downloads>
PS C:\Users\Shree priya\Downloads> c:: cd 'c:\Users\Shree priya\Downloads'; & 'c:\Users\Shree priya\AppData\Local\Microsoft\Windows\apps\python\python.exe' -c 'import sys; sys.path.append('c:\Users\Shree priya\Downloads\'); import aac1; aac1.reverse_string('Hello, World!')'
PS C:\Users\Shree priya\Downloads> ^C
```

Task – 3: Modular Design Using AI Assistance (String Reversal Using Functions)

Prompt :

Generate a code for Modular Design Using AI Assistance (String Reversal Using Functions)

Code:

```
# String Reversal Using Functions
def reverse_string(s):
    reversed_s = ""
    for char in s:
        reversed_s = char + reversed_s
    return reversed_s
input_string = "Hello, World!"
reversed_string = reverse_string(input_string)
print(reversed_string)
```

Output:

```
PS C:\Users\Shree priya\Downloads> c:: cd 'c:\Users\Shree priya\Downloads'; & 'c:\Users\Shree priya\AppData\Local\Microsoft\Windows\apps\python\python.exe' -c 'import sys; sys.path.append('c:\Users\Shree priya\Downloads\'); import aac1; aac1.reverse_string('Hello, World!')'
PS C:\Users\Shree priya\Downloads> ^C
```

Task – 4: Comparative Analysis – Procedural vs Modular Approach (With vs Without Functions)

Prompt :

Generate a code for Comparative Analysis – Procedural vs Modular Approach (With vs Without Functions)

Code :

```
# With vs Without Functions
def reverse_string(s):
    reversed_s = ""
    for char in s:
        reversed_s = char + reversed_s
    return reversed_s
input_string = "Hello, World!"
reversed_string = reverse_string(input_string)
print(reversed_string)
```

Output:

```
PS C:\Users\Shree priya\Downloads> c:: cd 'c:\Users\Shree priya\Downloads'; & 'c:\Users\Shree priya\AppData\Local\Microsoft\Windows\apps\python\python.exe' -c 'import sys; sys.path.append('c:\Users\Shree priya\Downloads\'); import aac1; aac1.reverse_string('Hello, World!')'
PS C:\Users\Shree priya\Downloads> ^C
```

Task – 5: AI-Generated Iterative vs Recursive Fibonacci Approaches (Different Algorithmic Approaches to String Reversal)

Prompt :

Generate a code for AI-Generated Iterative vs Recursive Fibonacci Approaches (Different Algorithmic Approaches to String Reversal)

Code :

```
# Different Algorithmic Approaches to String Reversal
# Using Slicing
input_string = "Hello, World!"
reversed_string = input_string[::-1]
print(reversed_string)

# Using reversed() Built-in
input_string = "Hello, World!"
reversed_string = "".join(reversed(input_string))
print(reversed_string)

# Using Recursion
def reverse_string_recursive(s):
    if len(s) == 0:
        return s
    return reverse_string_recursive(s[1:]) + s[0]

input_string = "Hello, World!"
reversed_string = reverse_string_recursive(input_string)
print(reversed_string)

# Using Stack
def reverse_string_stack(s):
    stack = [char for char in s]
    reversed_s = ""
    while stack:
        reversed_s += stack.pop()
    return reversed_s

input_string = "Hello, World!"
reversed_string = reverse_string_stack(input_string)
print(reversed_string)
```

Output:

```
PS C:\Users\Shree\Downloads> cd "C:\Users\Shree\Downloads" & python3 "C:\Users\Shree\Downloads\different_approaches_to_string_reversal.py"
Hello, World!
Hello, World!
Hello, World!
Hello, World!
PS C:\Users\Shree\Downloads>
```